

Week 1 Reflection
TCSS-498 Directed Reading (Raghavi Sakpal)
Seth Campbell

The Denavit-Hartenberg (DH) convention is a method used for systematically describing and planning robotic motion. It treats each joint in the assembly as a link that adjusts and defines the orientation of each subsequent link's position in space. The base of the chain is the fixed point (usually the base or mounting point) and the terminal position is the end-effector (usually a gripping mechanism). For each coordinate frame in the assembly, 4 parameters are used to establish a relationship between two adjacent frames: link length, link twist, link offset, and joint angle. Link length, twist, and offset are static values based on the physical geometry of the robot and remain constant. The joint angle is the single variable representing motion for a revolute joint.

The DH convention has two geometric requirements when establishing the relationship between two adjacent frames: the x-axis of each frame must be perpendicular to the previous frame's z-axis, AND it must intersect the previous frame's z-axis. Once the coordinate frames are established, a transformation matrix is built for each joint and multiplied sequentially from base to end-effector to determine the manipulator's final position and orientation in space.

This chapter largely addresses revolute joints, like those driven by servos and stepper motors, and how each joint's coordinate frame is aligned. The z-axis defines the axis of rotation and for a servo motor, this is the axis the shaft revolves around. The x-axis is the common normal, pointing along the shortest perpendicular distance between the current and previous joint's z-axis. The y-axis is the cross product of the two and is orthogonal to both the z and x axes, determined by the right-hand rule. Because I will be using C++ for this project, matrix multiplication will not be available in the standard library and a basic function to handle this will need to be written to support the forward (and later inverse) kinematic chain. There is a shortcut I describe in the next paragraph that I will also implement to reduce the overall computation load, since real-time response requires frequent sampling and the overhead adds up quickly.

This chapter has given me a lot of insight into the movements of the follower arm (teleoperated), which I currently see using forward kinematics to match the inverse kinematics of the leader arm. One useful outcome of the DH method is that it is possible to derive a single 'consolidated' matrix containing shorthand equations for all required movements. These derived matrices are readily available online, and after I complete an implementation for the method described above, I will then implement this 'consolidated' version of it to cut down on computation. I do see value in performing this derivation myself, or at least diving into a step-by-step explanation to better understand the math behind this, but I also believe it's outside the scope of the directed reading.

Since my setup is using direct drive servo motors, meaning no reducer or gear assembly is utilized for any joints, the translation between leader and follower will be a direct 1-1. This also means I only have to be concerned with revolute joints, although for later applications understanding how other joints operate would be crucial. The connection between joints will also be aligned along the x-axis between the joints meaning the link twist and link offset values will be 0 across the entire assembly, simplifying the calculations slightly. Currently parts are printed and I'm standing by for the delivery of the motors used in this setup. As a rule of thumb, I assume minor complications on assembly, but nothing that will delay the project timeline.